CognoDB Graph Database Benchmark — Final Performance Report

Benchmark scope: CognoDB, Neo4j, Memgraph, FalkorDB, and TigerGraph. The supplied benchmark results contain seven workloads, 100 measured runs for read workloads, and 40 operations for mixed read/write with four workers.

Executive Summary

FalkorDB is the strongest performer across the seven read-oriented workloads in this benchmark, achieving the lowest mean latency and highest throughput in every one of those workloads. Neo4j is the strongest result for the mixed read/write workload. CognoDB is faster than TigerGraph on average, but remains behind FalkorDB, Neo4j, and Memgraph in the measured environment. These results are environment-specific benchmark observations rather than universal database rankings.

Overall Results

Database	Avg mean latency (ms)	Avg P95 latency (ms)	Avg throughput (ops/s)
FalkorDB	53.129	157.861	28.537
Neo4j	68.279	112.178	20.502
Memgraph	163.738	270.018	8.180
CognoDB	286.816	524.032	4.765
TigerGraph	382.016	575.831	4.040

Workload Winners

Workload	Latency winner	Mean latency (ms)	Throughput winner	Throughput (ops/s)
Point lookup	FalkorDB	27.856	FalkorDB	35.893
1-hop traversal	FalkorDB	51.265	FalkorDB	19.505
2-hop traversal	FalkorDB	25.895	FalkorDB	38.611
3-hop traversal	FalkorDB	47.681	FalkorDB	20.971
Filtered lookup	FalkorDB	50.708	FalkorDB	19.719
Aggregation	FalkorDB	26.821	FalkorDB	37.279
Mixed read/write	Neo4j	68.759	Neo4j	55.111

Interpretation

FalkorDB: Best overall read/query performance in the supplied measurements. Its particularly low 2-hop latency (25.895 ms mean) and high throughput (38.611 ops/s) are notable.

Neo4j: Strong low-latency and throughput results, and the best mixed read/write result at 68.759 ms mean latency and 55.111 ops/s.

Memgraph: Middle-tier performance across the read workloads, with relatively consistent latency and a substantially better result than CognoDB and TigerGraph in this test environment.

CognoDB: Mean latency is lower than TigerGraph across the aggregate workload set, but the current measurements show a sizable performance gap versus FalkorDB and Neo4j.

TigerGraph: The server-side 3-hop implementation substantially avoided the earlier runaway client-side behavior, producing a completed 3-hop benchmark at 545.735 ms mean latency. Its 2-hop workload remained expensive at 601.449 ms mean latency.

Methodology and Caveats

The comparison uses the benchmark CSV results supplied in the project. Metrics include minimum, maximum, mean, P50, P95, and throughput. The workloads were run with a fixed random seed of 42; read workloads used 10 warm-up runs and 100 measured runs, while mixed read/write used four workers and 40 total operations (20 reads and 20 writes). The benchmark is therefore useful for comparative analysis under the tested setup, but hardware, network, deployment mode, indexing, query formulation, caching, and database configuration can materially affect absolute performance.